Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Automated Single Pathogen Tracking for Rapid Diagnosis of Antimicrobial Drug Resistance (Infectious Disease)**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Advanced Bio		**Advanced Bio**	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	**Biosignal Measurement / Diagnostic Devices**	**Medical Bio**	**Reagents / Diagnostic Reagents**
Concept of Proposed Technology	**Main Need:** Drug-resistant pathogens pose an increasing threat to healthcare systems worldwide. The World Health Organization (WHO) publishes a list of existing and emerging drug-resistant pathogens and has described new, rapid diagnostic technologies as a major need. Rapid treatment with an effective antimicrobial is critical to positive clinical outcomes following infection. However, prescribing the correct drug requires knowledge about the drug resistance profiles of the presumed pathogens. Traditional culture-based methods of establishing resistance profiles can take up to 72 hours, resulting in the patient receiving either too narrow- or broad-spectrum drug treatments. Technologies that provide more rapid drug resistance profiles would improve patient outcomes, and aid in the world-wide combat against emerging drug-resistant pathogens. However, no current rapid diagnostic technology has provided the diagnosis using devices/techniques that meet the time-to-answer, ease of use, and cost-of-test requirements for true commercialization and integration into healthcare workflows. **Core Concept:** Our team has developed a technology to directly capture microbial pathogens from clinical samples (including whole blood) and then perform cell-by-cell assessments of drug susceptibility without the need for growth and culture of the captured pathogens. The entire sample-to-diagnosis process is performed in less than two hours and can be used to assess the effectiveness of multiple drugs on the limited number of pathogens (i.e., 1-10 CFU per mL of blood sample).			

Importantly, our platform is not limited to measuring specific drug resistance genes and can be used broadly against all microbial pathogens.

Main Functions:

1) **Automated Isolation of Pathogens (from blood)** – Centrifugation has been a mainstay of microbiology for decades; however, its use in rapid diagnostics and clinical workflows remains limited due to the difficult hands-on manual steps involved with pathogen isolation using centrifugation methods. Our team has developed the first automated centrifugation system for microbial isolation, which utilizes a novel switch built into the low-cost centrifuge tube disposable to automatically protect the sample pellet and isolate the bacteria during reagent additions. This system enables hands-free separation of bacterial pathogens from clinical samples (including blood), even at low concentrations (i.e., <1-10 CFU/mL), and isolation in low (i.e., 50 microliters or less) testing volumes.
2) **Deterministic Pathogen Tracking**
   Our group has specialized in capturing isolated microbes in digital liquid droplets or chambers, so that each individual microbe can be monitored and the effect of a drug on the microbes tracked (without the need for long growth and culturing steps). However, in the past these systems required use of expensive microfluidic devices and could only be used to assess a small number of drugs on the available microbial sample (while clinical sites typically required screening of 3–9 drugs). Our team has developed a new microbe trapping system that enables deterministic trapping of each bacterium collected from the sample isolation system (Main Function #1), tracking each microbe over time, and rapid introduction and screening of multiple drugs over each capture bacteria.
3) **Direct RNA Analysis for Rapid ID and Antimicrobial Susceptibility Testing (AST)**
   Recently, our group (and partners) have shown that use of RNA-based probes enables simultaneous identification and AST assessment in 15–60 minutes following drug incubation (across many microbe species/strain). Our probes target sequences that contain both conserved and species-specific RNA sequences and enable universal screening and identification of any microbe present in the sample. In addition, the reagent/probes target key RNA molecules that are rapidly affected by the presence of broad categories of antimicrobial drugs and demonstrate rapid changes in concentration during short incubation periods.

Potential Uses:

Our infectious disease diagnostic system overcomes the current limitations of microbial identification and AST devices and methods by (1) using unique molecular probes to perform rapid RNA-response assays for rapid phenotypical AST/drug profiling and (2) eliminating the need to culture microbes for clinical diagnosis through use of proprietary microbial isolation, separation, and tracking systems. This system can be used to replace the current slow AST tests used in clinical settings worldwide, provide rapid assessment of microbial drug resistance, and enable clinicians to detect drug resistance early enough to prescribe personalized/correct drug treatments for infected patients. Our technology is a collaborative opportunity for either medical device or reagent provider SMEs looking for novel devices for next-generation diagnostic capabilities or new reagent products (coupled to a novel device

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	and medical application).
Rationale for Joint R&D Project	Our academic research team has brought prototype devices/reagents to TRL3 level development and is now ready to collaborate with SMEs in the medical/diagnostic reagent and/or medical device space. This collaboration is essential as the team moves from testing spiked blood samples to performing tests within clinical operational environments (i.e., movement to TRL7). Work with the SMEs will enable design-for-manufacturing and clinical testing with devices fully vetted with manufacturing and distribution partners. In addition, reagent provider SMEs will enable use and testing using kits assembled within commercialization and regulatory testing facilities.
Key Words	1. Infectious Disease 2. Drug Resistance 3. Microbial Identification (ID) 4. Antimicrobial Susceptibility Testing (AST) 5. Rapid Diagnostics

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)
R&D Objectives	In our TIPA SME partnership, our team can work with either a device or reagent company to further develop the key component technologies enabling rapid infectious disease (i.e., drug resistance) diagnosis. Our current team has internal capabilities to aid either a device- or reagent-based Koren SME (i.e., our team can handle either device or reagent commercialization challenges) but seeks a partner in one of these areas for co-development and commercialization. Our primary objectives across the whole team will be: Phase I - 1) Extend the current prototype (TRL 3) sample preparation components (i.e., automated/smart centrifuge tubes and system) to a final design using injection molding and commercial-ready components (collaborative design for manufacturing completion) 2) Extend the rapid bacteria/RNA-analysis optical detection system to final commercial design and test samples in fieldable/clinical user ready unit (collaborative design for manufacturing completion) 3) Finalize reagent components (kit), including manufacturing site and partner 4) Complete commercialization plan with SME partner, including Phase II testing plan (RPI has available clinical and manufacturing partners but is open to SME taking on any of these strategic roles, as needed) Phase II - 1) Manufacture Alpha prototype using collaborative SME/RPI designed systems and components 2) Run pilot fabrication of final sample preparation components and reagent kits 3) Install system into clinical partner/field, including system, consumables, and reagents 4) Perform pilot clinical test validating equivalent microbial identification and drug susceptibility detection compared to standard technology (with sample to answer time of <1 hour compared to 0.5–3 days using standard tech)			
R&D Target Specifications	Bacterial Identification and Drug Resistance Specification: Sample preparation components - Functions for blood, swab, and sputum samples - Functions for samples from <0.5 mL to >10 mL - Function for all bacteria concentrations (including <1 CFU/mL blood infections) RNA-reagents and Detection System - Universal identification for all infectious microbes/bacteria (single reagent stream) - Quantitative assessment of microbial load/concentration - Rapid identification of drug resistance (<1 hour) for choice of 3-5 drugs (user choice/input) Clinical Readout – - Description of all microbes present in the sample - Drug resistance status and breakpoint for tested drugs (for each identified microbe)			

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R&D Plan	**Phase I of TIPA Program** **Engineering Plan** The RPI team TRL3 system will be reviewed by the SME partnership team (including desk-top system, consumable cartridges, and reagents). RPI and SME partners will complete design-for-manufacturing assessments, using either current RPI manufacturing partners available or SME's internal or external choice of manufacturing partners. Engineering Deliverable By the end of phase I, the combined team will have completed design for manufacturing analysis (e.g., injection molding design for all consumables, final parts list for commercial-scale RNA-bacteria reader). RPI can rapidly prototype all design parts and perform validation tests using bacteria/blood samples during this process. Engineering Milestone/Goal Completion of design for manufacturing with identified go-to-market partners, and validation tests of equivalence with current RPI TRL3 prototypes (i.e., sample to answer time, bacteria collection efficiency, and RNA bacteria identification times). **Commercialization Plan** The team will determine commercial/go-to-market strategy on final bacteria/infection targets (i.e. first clinical sample and pathogen types). The team will finalize commercial plan, including cost-value assessment on cost per test (enabled by healthcare system and market) and total market capitalization available for initial/go-to-market test kits. This market data will be integrated into the engineering plan to ensure that test costs per kit are acceptable to the diagnostic market, and the value-add of rapid drug resistance infection analysis is re-couped in commercial launch. Commercialization Deliverable Market assessment and test kit down selection (i.e., select first sample/microbial target for product launch). Commercialization Milestone/Goal Delivery of design for commercialization reagent kit and consumables that meet target costs/price per test. **Clinical Testing Plan** The short phase I period will not require clinical testing (which will be performed in phase II should we be selected). However, the RPI team will test the final prototype systems designed for manufacturing/commercialization with the partner SME team using in-house blood/infected samples. This will prove equivalent to the current RPI TRL 3 prototype and enable phase II clinical testing at a field/hospital site. The team will also work with a hospital partner to establish the phase II plan for testing, including the number and type of samples testing and the control/standard assay run in parallel with the rapid test.

	Clinical Testing Deliverable Down-selected clinical testing partner and site, and formalized testing plan including sample types, control and gold-standard parallel tests required. Full timeline, cost, and risk mitigation plan from clinical partner site. Clinical Testing Milestone/Goal Integrated testing plan for phase II work, including production site/pilot manufacturing plan for consumables and reagents, and clinical testing plan at hospital partner site. **Phase II of TIPA Program** The goal of phase I will be to finalize the pilot production and clinical testing plan with our SME partner based on both clinical and business/market feedback. The RPI team has internal capability for several important items, including: - Optional clinical testing facility near RPI site - Manufacturing partners for either RNA optical analyzer system, consumables/cartridges, and/or reagent kits - Initial data on US market clinical sample drug/drug profiles most in need (i.e., U.S. domestic market analysis) - Current TRL 3 prototype systems for sample preparation and bacteria analysis In phase I, the RPI team is flexible to use SME expertise and facilities/capabilities for any or all of the above necessary go-to-market components/needs. Phase I will be about making a team plan based on the sum of RPI team and SME capabilities. In phase II, this combined knowledge and plan will enable a full clinical test which will assess quantification of the limit of detection of the new rapid microbial/drug resistance assay, the speed of sample-to-answer using clinical samples (compared to state of the art), and the categorical agreement of the rapid tests compared to the slow/days-long gold standard methods.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The RPI led team has TRL 3 level prototypes for applying/testing within this TIPA program, and intellectual property (pending) for the novel sample preparation and analysis system and tools. In addition, the current RPI team has knowledge and plan that will enable a full clinical test, which will assess quantification of the limit of detection of the new rapid microbial/drug resistance assay, the speed of sample-to-answer, and categorical agreement of the rapid drug resistance results to the slow gold-standard/current approaches. RPI itself also has prototype and design/fabrication capabilities and blood/clinical sample prototype testing available, as needed.
Expected R&R of Korean SME Partner	Due to the internal RPI team's readiness and capabilities, we have very flexible SME partner possibilities and are very flexible on desired partner/fit and responsibilities. SME partners could include: - Optical-based medical diagnostic device companies seeking novel products in single cell capture and analysis and/or infectious disease diagnostics - Medical diagnostic device companies seeking potential additional revenue sources on reagents components - Reagent component and kit companies, seeking device partners to generate reagent sales revenues - Medical device partners within the Korean healthcare space that desire interactions with U.S.-based hospital and testing partners (and markets) - SMEs with general device or medical experience that would like to take advantage of our many capabilities on all aspects of commercialization, including device, cartridge, reagent, and testing partners

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	At the end of both phases of a TIPA program, our SME partner will have completed a pivotal clinical trial showing that our rapid microbial identification and drug resistance assessment system can provide drug resistance information to clinicians in approximately one hour (compared to many hours to days that it takes current systems). This will enable medical professionals to prescribe the correct medication to patients and eliminate giving the wrong medication to patients that harbor a drug- resistant infection. Commercialization of this system (which would be enabled by the phase II clinical test) would result in the patients no longer receiving too narrow- or too broad-spectrum drug treatments. These more rapid drug resistance profiles would improve patient outcomes, and aid in the world-wide combat against emerging drug-resistant pathogens. No current rapid diagnostic technology has provided this type of diagnosis using devices/techniques that meet the time-to-answer, ease of use, and cost-of-test requirements for true commercialization and integration into healthcare workflows. Our project would solve these challenges and fill this void.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization